

❏ What is DNS?

DNS (Domain Name System) resolves human-readable domain names (e.g., `www.example.com`) into IP addresses (e.g., `192.0.2.1`). It functions like a phonebook for the internet.

❏ Components of DNS

Component	Role
Resolver	Client making DNS queries
DNS Server	Responds to queries (authoritative or caching)
Zone File	Stores domain data
Records (A, CNAME, MX, etc.)	Define mappings
Bind/Named	Common DNS server software on Linux

📄 DNS Client Configuration

On **client-side**, DNS is configured by:

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/etc/resolv.conf

bash

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```
nameserver 8.8.8.8 nameserver 1.1.1.1
```

(Specifies DNS servers to use)

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/etc/hosts

Local static mapping of names to IPs:

bash

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```
127.0.0.1 localhost 192.168.1.10 server.example.com server
```

❏ DNS Server Configuration (Using BIND)

❏ Key BIND Configuration Files:

File	Purpose
/etc/named.conf	Main config file
/var/named/db.example.com	Forward zone file
/var/named/db.1.168.192.in-addr.arpa	Reverse zone file

☒ Steps to Set Up a DNS Server (BIND)

1.

Install BIND:

bash

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```
sudo apt install bind9    # Debian/Ubuntu sudo yum install bind    # RHEL/CentOS
```

2.

Edit named.conf:

bash

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```
options { directory "/var/named"; allow-query { any; }; recursion no; }; include "/etc/named.rfc1912.zones";
```

3.

Create Forward Zone (db.example.com)

bash

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```
$TTL 86400 @ IN SOA ns1.example.com. admin.example.com. ( 2025061901 ; Serial 3600 ; Refresh 1800 ; Retry 604800 ; Expire 86400 ) ; Minimum IN NS ns1.example.com. ns1 IN A 192.168.1.1 www IN A 192.168.1.100
```

4.

Create Reverse Zone (db.1.168.192.in-addr.arpa)

bash

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```
@ IN SOA ns1.example.com. admin.example.com. ( 2025061901 ; Serial 3600      ; Refresh 1800      ; Retry 604800      ; Expire 86400 ) ;
Minimum IN NS ns1.example.com. 100 IN PTR www.example.com.
```

5.

Configure Zone in /etc/named.conf:

bash

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```
zone "example.com" IN { type master; file "/var/named/db.example.com"; }; zone "1.168.192.in-addr.arpa" IN { type master; file "/var/named/db.1.168.192.in-addr.arpa"; };
```

6.

Start and Enable DNS Service:

bash

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```
sudo systemctl start named sudo systemctl enable named
```

7.

Verify DNS Configuration:

```
bash
```

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```
named-checkconf named-checkzone example.com /var/named/db.example.com dig @localhost example.com nslookup example.com localhost
```

☒ Common DNS Record Types

Record	Purpose
A	Maps name to IPv4 address

AAAA	Maps name to IPv6 address
MX	Mail exchange (mail server)
CNAME	Canonical name (alias)
NS	Name server
PTR	Reverse DNS lookup
TXT	Text information

☒ DNS Security

- **DNSSEC:** Prevents spoofing
- **Rate Limiting:** Prevents DNS amplification attacks
- **Allow-query & recursion:** To restrict abuse